

PRISMA 2020 Checklist for Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection: SLR

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title p.1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Section I (Introduction) & Section II (Related Work)
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Section III.A (Planning the review) & Section III.B (Research questions)
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Section III.D (Inclusion criteria), Section III.E (Exclusion criteria), & Section III.I (Data synthesis)
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Section III.C (Information Sources) & Table III
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Section III.C (Information sources) & Table III
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Section III.F (Study selection)
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Section III.H (Data extraction and analysis)
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Section III.H/I (Data extraction and analysis)
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Section III.H/I (Data extraction and analysis)
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Section III.G (Quality Assessment) & Section III.H (Data extraction and analysis)
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	N/A: No effect measures were used because this

PRISMA 2020 Checklist for Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection: SLR

Section and Topic	Item #	Checklist item	Location where item is reported
			was a qualitative/descriptive SLR
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Section III.I (Data Synthesis) & Section V (Results)
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Section III.H (Data Extraction), Section III.I (Data Synthesis), and Section V.C / Tables IX–XII
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Section IV (Bibliometric Analysis) & Section V (Results, Tables V–XIII)
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Section V (Results, Introductory paragraph)
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Section V.D (RQ4: Related Challenges Affecting Reliability, Reproducibility, and Generalizability)
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	N/A (Qualitative narrative synthesis)
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Not formally assessed; related issues are discussed through QA5, reproducibility limitations, Section III.G, Section V.D, and Section VII.
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not formally assessed; evidence confidence was addressed through the QA threshold in Section III.G and limitations in Section VII.
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Section III.F & Fig. 1 (PRISMA flow diagram)
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Section III.F plus Supplementary Folder 03: Table of Excluded

PRISMA 2020 Checklist for Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection: SLR

Section and Topic	Item #	Checklist item	Location where item is reported
			Studies / QA exclusion justification.
Study characteristics	17	Cite each included study and present its characteristics.	Results Section V p. 8-17
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Section III.G and Supplementary Folder 03: QA_Scoring_Master_List / excluded-study QA table.
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	N/A for effect estimates; individual study characteristics are summarized in Section V Tables V–XIII and Supplementary Folder 04.
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Section V (Results) & Section III.G (Quality Assessment)
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A (Qualitative narrative synthesis)
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Section V.D (RQ4: Related Challenges)
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A (Qualitative narrative synthesis)
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Not formally assessed; discussed under QA/reproducibility limitations in Section III.G, Section V.D, and Section VII.
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not formally assessed using a formal certainty framework; addressed through QA scoring and limitations.
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Section V (Results) &

PRISMA 2020 Checklist for Datasets and Data Preprocessing Techniques for Machine Learning-Based Insider Threat Detection: SLR

Section and Topic	Item #	Checklist item	Location where item is reported
			Section VI (Discussion)
	23b	Discuss any limitations of the evidence included in the review.	Section V.D (Challenges) & Section VII (Limitations)
	23c	Discuss any limitations of the review processes used.	Section VII (Limitations)
	23d	Discuss implications of the results for practice, policy, and future research.	Section VI (Discussion) & Section VIII (Conclusion)
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Declaration (p.17)
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Declaration (p.17)
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Declaration (p.17)
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Declaration (p.17)
Competing interests	26	Declare any competing interests of review authors.	Declaration (p.17)
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Supplementary Folders 01–06 and Data Availability statement 17

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71. This work is licensed under CC BY 4.0. To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>